

US Domestic Flights 2015 — Whose Delay Is It?

1. Scope of the assessment.

The report answers a single question: how much of an airline's arrival delay is its own responsibility, and how much belongs to the network it operates in.

2. Architecture

Data exploration was done in Python; the model was built in Power BI, with ingestion, transformation, and semantic layers separated into four query groups. Staging and transformation queries have load disabled, so only the four model tables materialize.

Group	Layer	Contents
00_Param	Configuration	pDataPath — the source folder, parameterised
01_Stage	Ingestion	The three source files, read with explicit types
02_Clean	Transformation	fnNormalizeAirport, Clean_Flights
03_Model	Semantic model	Fact_Flights + Dim_Date, Dim_Airline, Dim_Airport

The model is a star schema: one fact table (Fact_Flights, grain = one scheduled flight) and three dimensions (Date, Airline, Airport), related one-to-many. All row-level attributes are derived in Power Query; DAX holds only measures.

3. Data quality

The source file contains four data quality issues. None raises an error on load, which is what makes them worth documenting. The data was first reviewed in Python, and that's where these issues were discovered.

3.1 Inconsistent airport coding.

During October (**486.165 flights, 8,35% of the dataset**) the origin and destination columns carry the DOT numeric identifier instead of the IATA code. Which means that all October data does not follow the same format as all the other months. The effect comes as the model recognises 628 distinct origin airports where there are in fact 322. This was resolved with a DOT-to-IATA mapping table built from public master data.

3.2 Times stored as HHMM integers

Time columns are integers that cannot be sorted or subtracted as times, and include the value 2400. To fix this, they are normalised to an hour of day between 0 and 23.

3.3 Heterogeneous denominators

Cancelled and diverted flights have no recorded arrival delay. Status flags were computed in Power Query so delay metrics run over the 5,714,008 completed flights and cancellation rate over the 5,819,079 scheduled ones. The states are mutually exclusive.

4. Benchmark Methodology (Mix-Adjusted)

Ranking airlines by raw average delay is misleading: an airline operating in congested winter hubs will look worse than one flying short, uncongested routes, regardless of actual management quality. Raw rankings measure where an airline flies, not how well it operates.

To correct for this, I built a mix-adjusted benchmark: for each airline, I calculate the delay expected given its actual mix of airports and months, using the industry-wide average in those same cells as the reference. The gap between actual and expected delay isolates genuine operational performance from the effect of route and seasonal mix.

This produced a concrete finding: Hawaiian Airlines ranks 3rd by raw average delay but drops to 9th once adjusted for its favorable route mix (short inter-island routes, no winter weather). US Airways shows the opposite pattern, climbing from 5th to 3rd. The methodology was independently validated by recalculating the same benchmark in Python (pandas), confirming the results across all 14 airlines.

5. Sources

The model is built on the 3 files provided in the advised kaggle URL (flights.csv, airlines.csv and airports.csv.)

However, the October fix is not possible with those files alone: airports.csv holds IATA codes only, and nothing in the supplied data identifies 10397 as Atlanta, so the mapping was built from two public sources — the official L_AIRPORT_ID lookup published by the Bureau of Transportation Statistics, and the open IATA code catalogue, which links that name to the three letter code.

6. How would this be built in production

In production I would keep the three layers, but separated so each one can be owned, tested and changed on its own:

- **Ingestion:** the source files land untouched in a storage area, so the original is always available for traceability.
- **Transformation:** cleansing, typing and airport code normalisation happen once, upstream of the report, and are written to clean tables with a stable structure.
- **Reporting:** the semantic model reads the clean tables and holds no cleansing logic.

Two operational points. At around 5.8 million rows a year, a refresh should reload only the recent data rather than the whole history. And the quality checks in particular the

coverage of the airport code fix should run automatically as part of the transformation, so a problem is reported by the process instead of being noticed in the report.

6. Limitations

- There is only one calendar year available, so year over year measures return blank by construction and none was included.
- Origin airport level only; adding destination needs a second, inactive relationship to the same dimension, activated with USERELATIONSHIP.
- The intraday chart covers the 05:00–23:00 operating day; 00:00–04:00 departures belong to the previous day's rotation and are excluded from it.